# Residential Rooftop Solar in India

## Approach Document

August 2024

solar square

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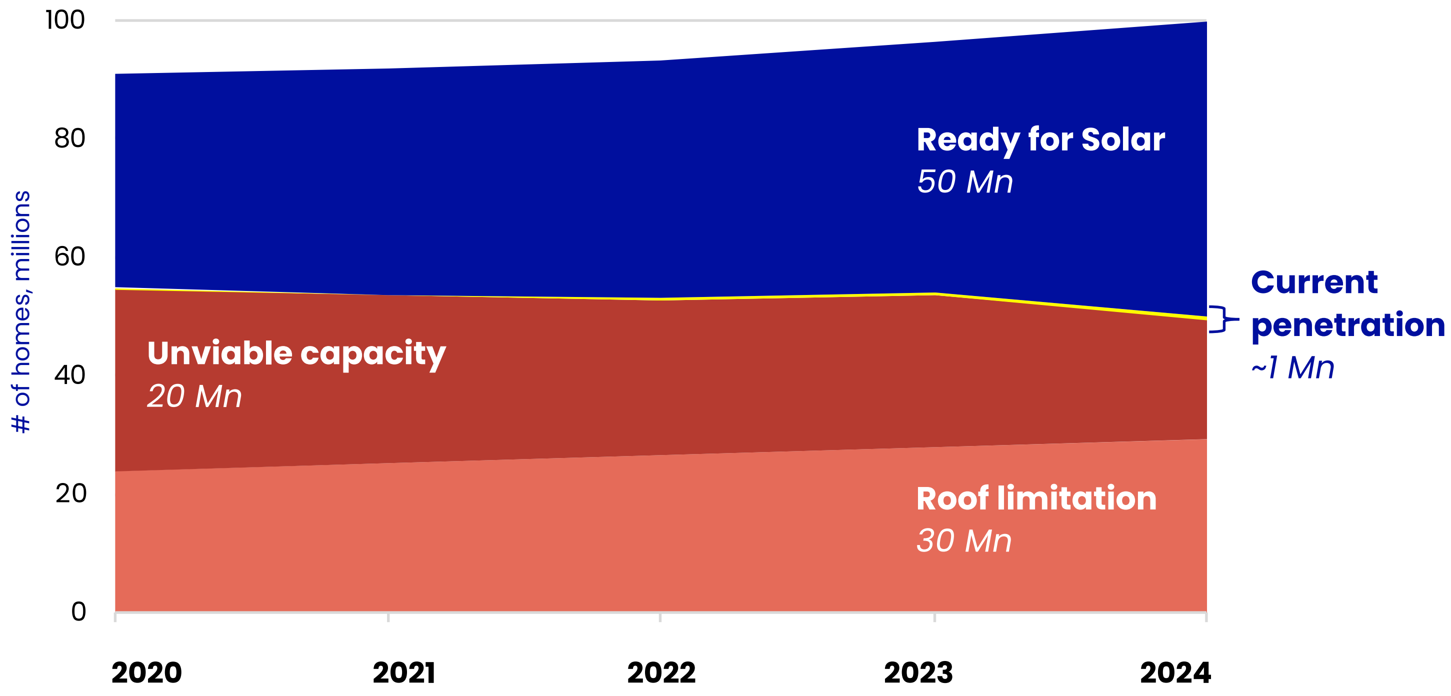
# India has 50 million homes ready for solarization…

**Estimated Rooftop Solar potential in India's Residential segment**
*Million homes*




**100 Mn homes**

Independent urban homes in India

**50 Mn homes**

Solar system economically viable

**~1 Mn**

Current Residential solar penetration in India

**20 Mn**

Household electricity consumption unviable (<200 units)

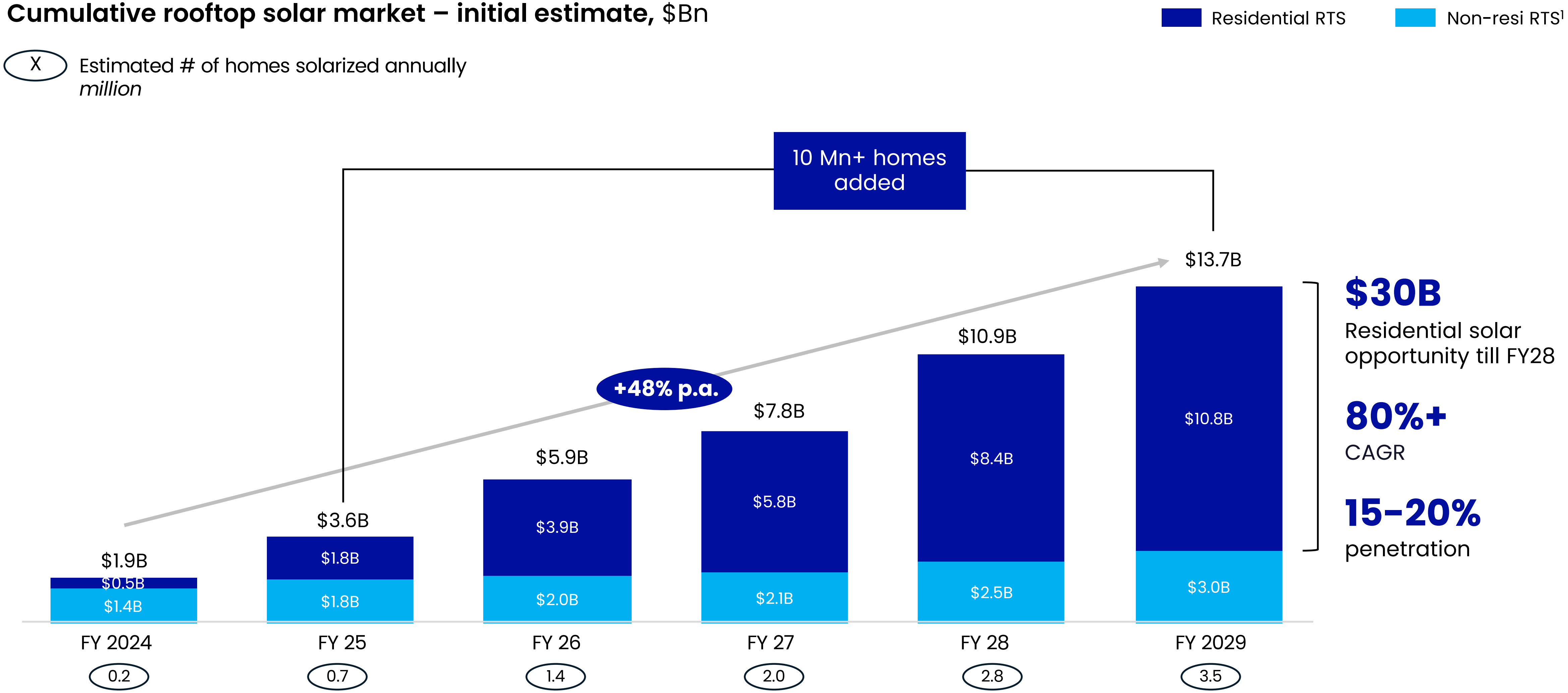
**30 Mn**

Rooftops unsuitable for Solar (shaded roofs, etc.)

Source: Census 2011, NSSO Survey, MNRE, Bridge To India

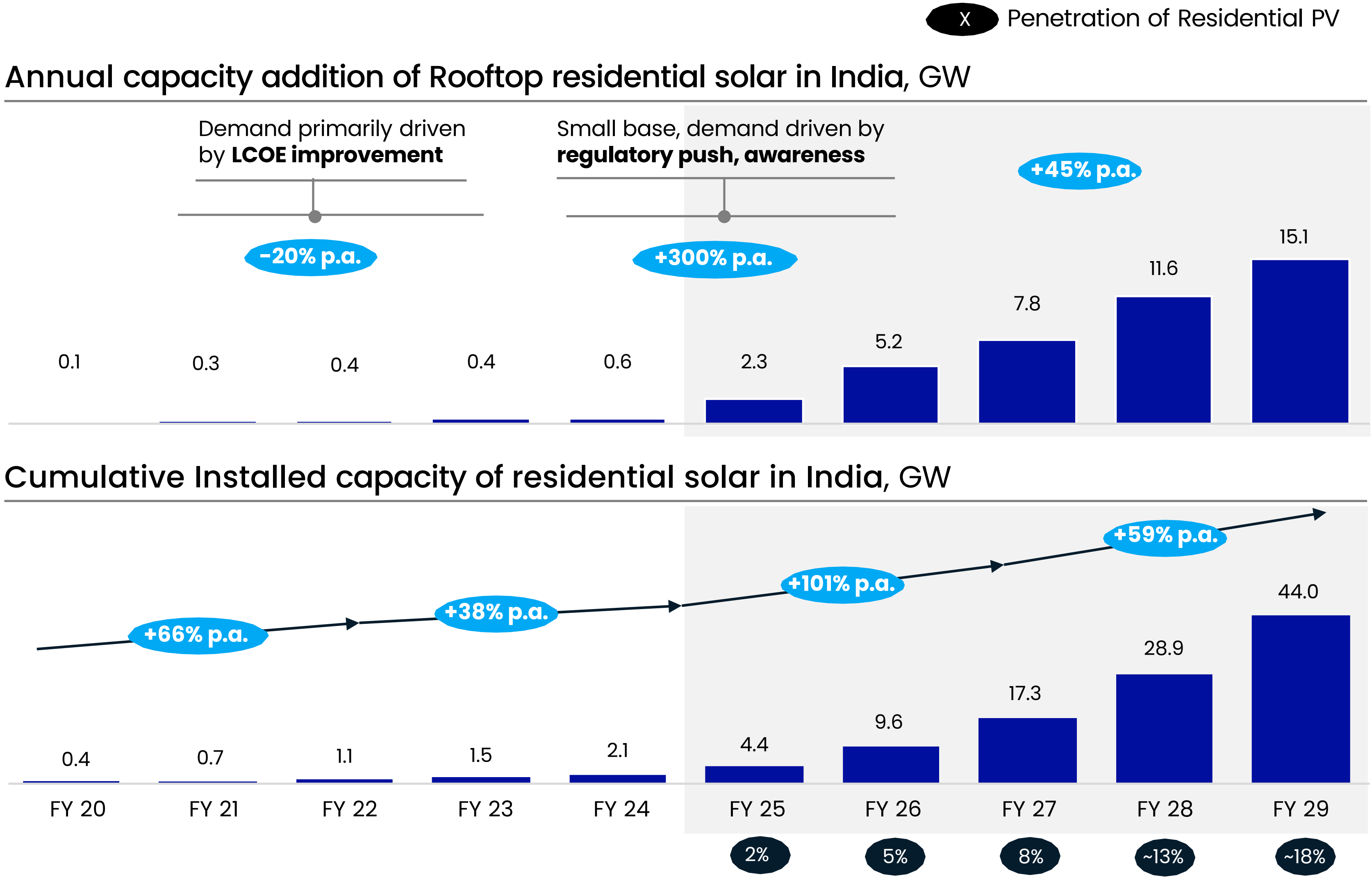
# ...and is expected to solarize 10 Million+ homes in the next 5 years, creating a $30 Bn+ opportunity in resi-solar

**Cumulative rooftop solar market – initial estimate,** $Bn




1. Commercial, industrial and government
Source: MNRE, Frost & Sullivan, McKinsey India power model

# Annual residential solar installations projected to continue growing at 80%+ CAGR through 2030




Source: MNRE, McKinsey India power model

## Key assumptions

**Installation capacity:** New players enter and existing players scale to bridge the forecasted supply shortage, some productivity improvements vs today

**Regulations:** Continued positive regulatory tailwind e.g., mandatory solar PV on certain residential new build, net metering policies expand to all states, single-window clearance

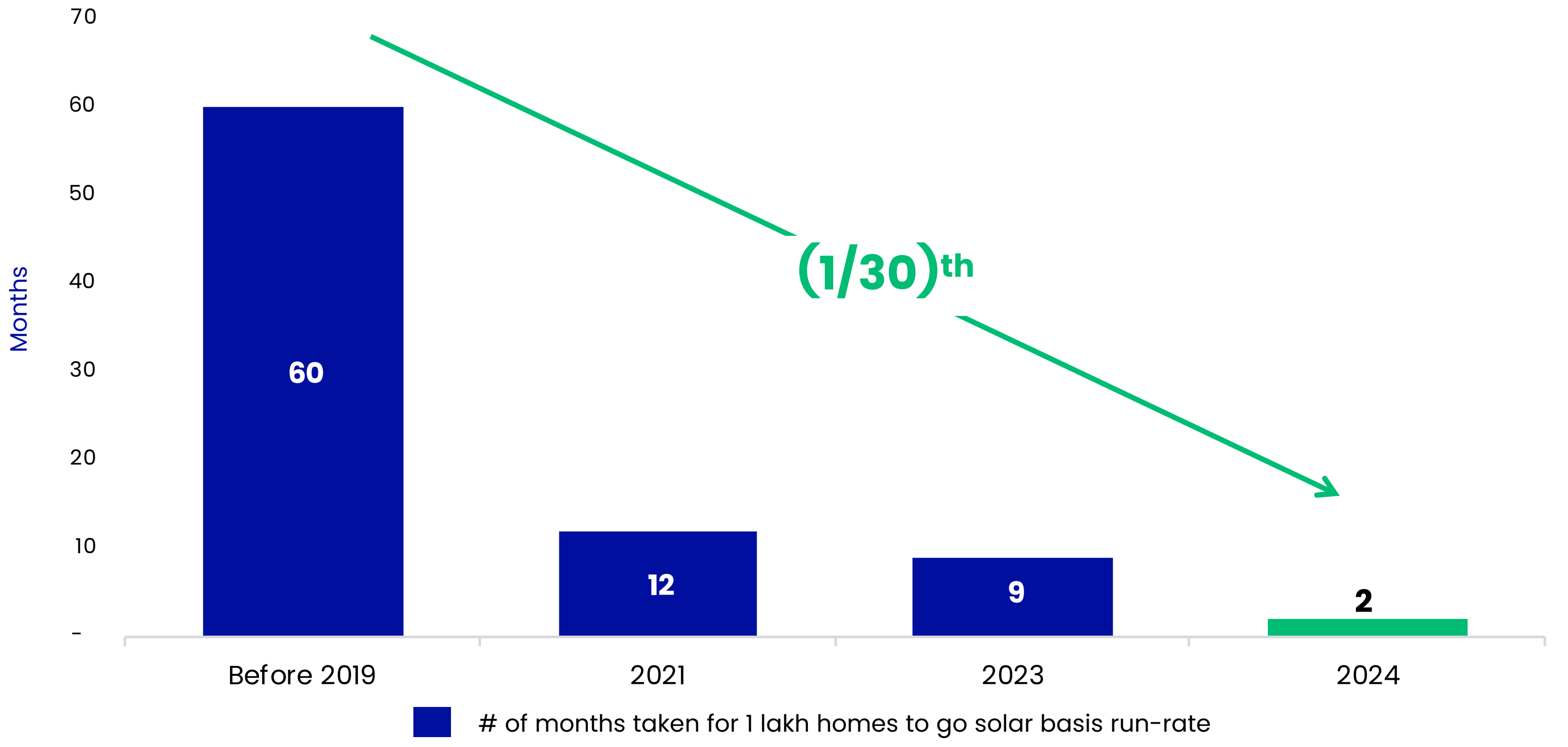
**Electricity prices:** Very modest or flat price development e.g., continue to increase in line with historic growth over last 3-4y of 4-5% p.a.

**Subsidies:** Continue with a gradual reduction

**Concluding comment:** India achieves ~60% of its FY2026 solar target of 40 GW capacity, primarily driven by resi-solar

# Market has inflected, with time taken for every 100K homes to go solar down from 60 months to 2 months

### Estimated time taken to add every 100k resi-solar customers in India
*in months*




### Key Takeaways

**~1 Mn**

Total # of homes solarized lifetime in India

**230k**

# of homes solarized in the last 5 months vs <1.5 Lakh homes solarized in all of 2023

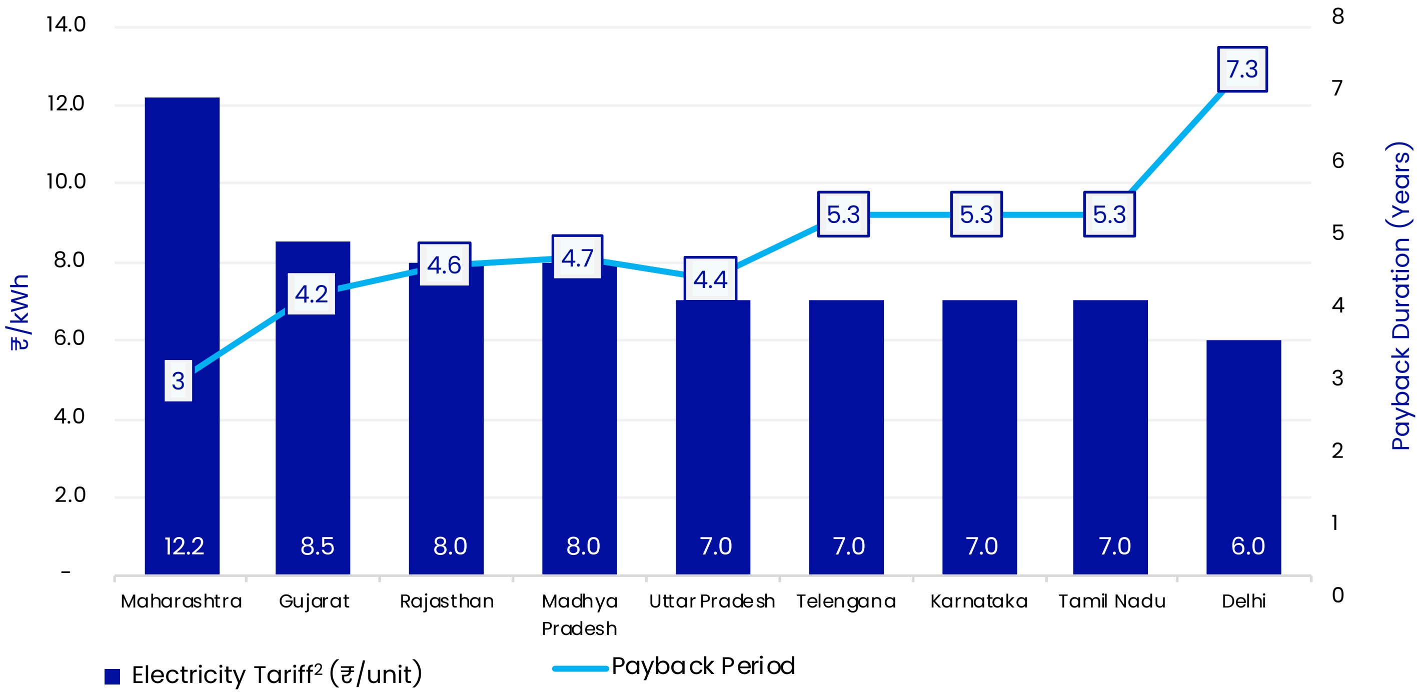
**<7 days**

Average time to receive solar permit from DISCOMs, down from 30+ days two years ago

Source: MNRE, Government of India

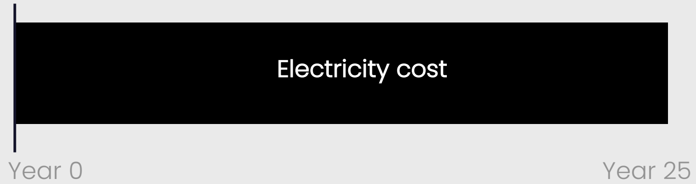
# **Going solar is a no-brainer for home-owners in India:**
Consumers breakeven in ~5 years and enjoy 20 years of free electricity

### Estimated payback period on a Solar PV system across states
*in months*




Simplified example

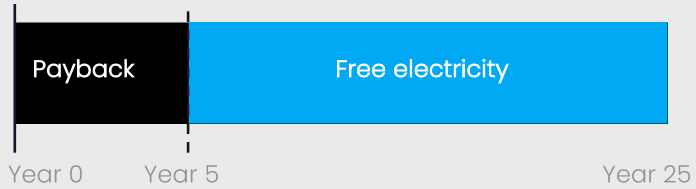
## Without Solar




**$12,000+ total outflow**

| Avg. tariff | X | Yearly consumption | X | Life of solar system |
|---|---|---|---|---|
| **$0.1/kWh** 3-5%Y-o-Y escalation | | **4200 units** | | **25 years** |

## With Solar PV System



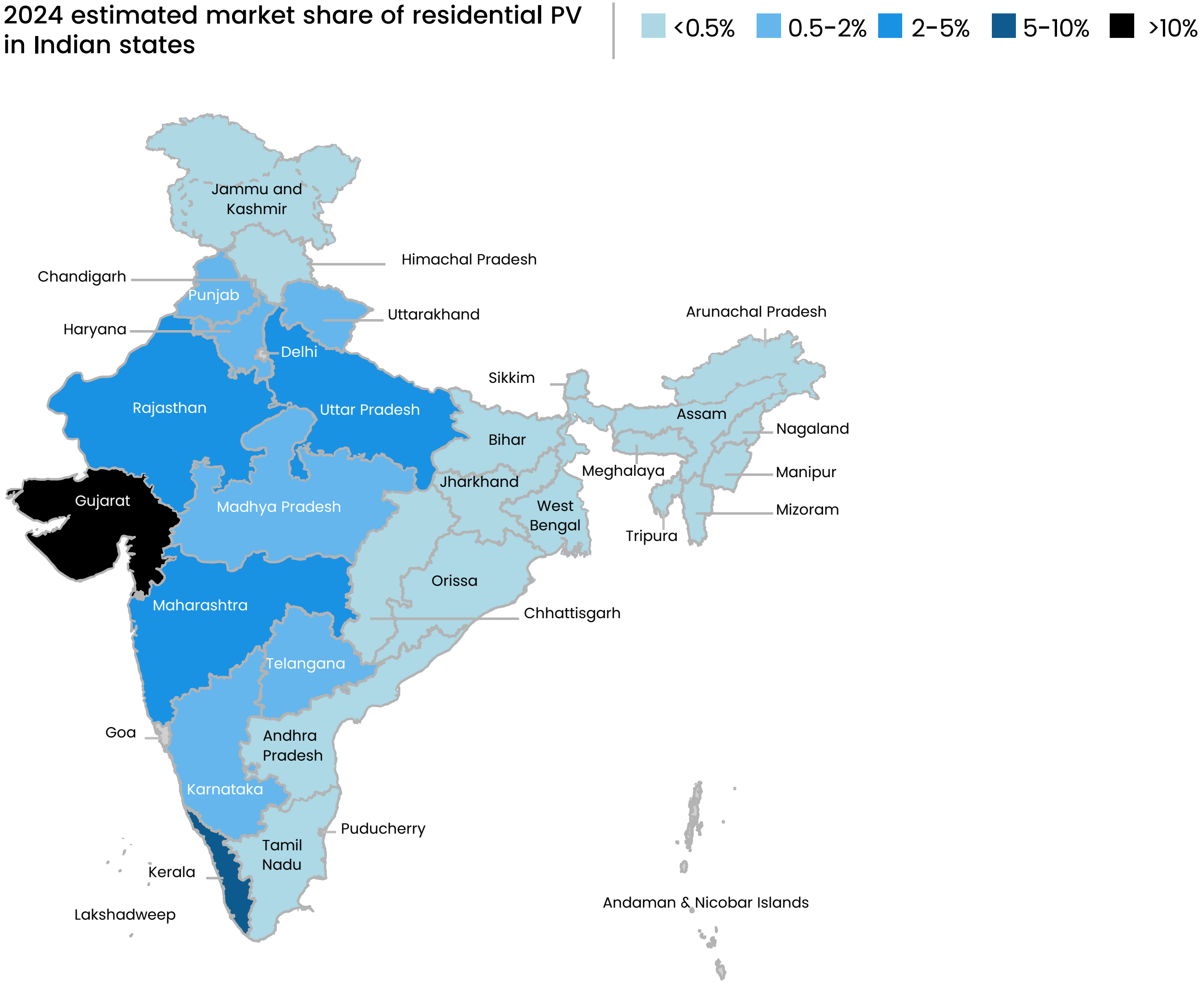

**$1,800 total outflow**

Cost of a 3kW system that generates 4200 units/ year

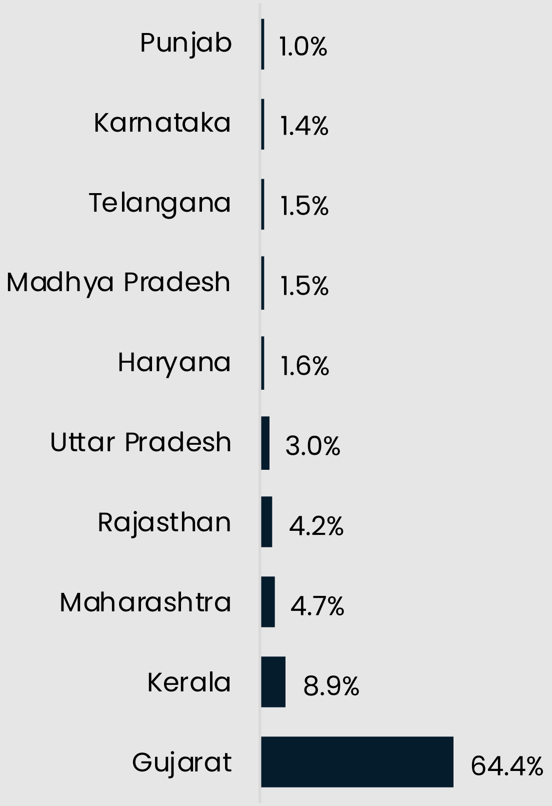
1. Assumed average cost per kW of $760 and Central Govt. subsidy of $580 on a 3kW system under PM Surya Ghar scheme
2. Avoided average electricity cost per unit basis Domestic Electricity LT Tariff Slabs of respective DISCOMs

# However, Gujarat is currently the only state with meaningful resi-solar penetration

2024 estimated market share of residential PV in Indian states



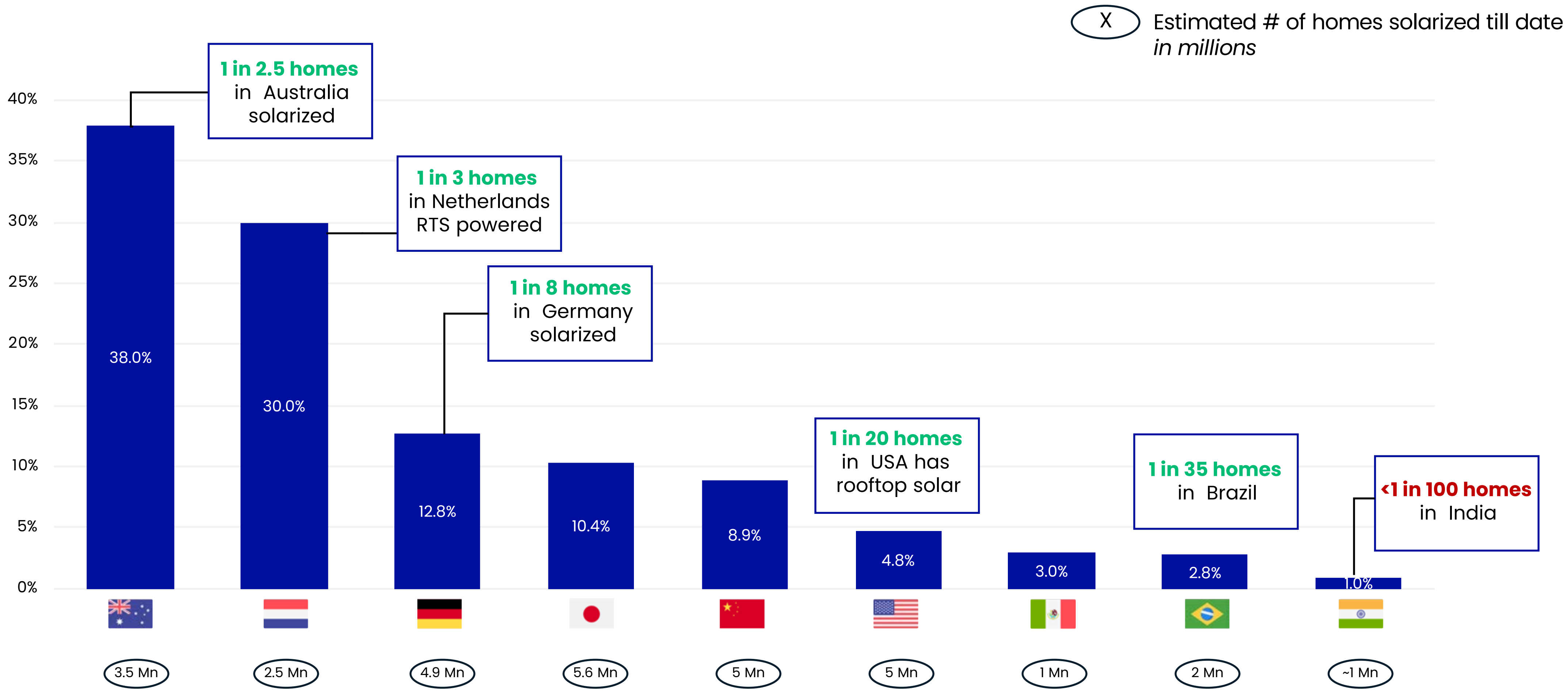

## Est. market share by state (%)




Source: MNRE, BridgeToIndia

# With only 1 in 100 homes solarized so far, India will be world's fastest growing resi-solar market this decade

## Residential rooftop solar penetration in key countries, # of homes




Source: PV Magazine, Bloomberg, IEA

# Large resi-solar outcomes have been created across the world…with marquee investors backing them

**List of countries**

**Companies**

Enpal.
1KOMMA5°
solfácil
octopus energy
goodleap
Project Solar
Palmetto
gosolr
M-KOPA
sunrun
vivint.Solar
niko
Brighte

SoftBank Group
TPG RISE CLIMATE
G2 VENTURE PARTNERS
SEQUOIA
Blackstone
ACCEL PARTNERS
QED INVESTORS
generation
Brookfield
lightrock
PICUS CAPITAL
ELEVATION
IFC
Investec
VALOR CAPITAL GROUP, LLC
EQT VENTURES
Goldman Sachs
MACQUARIE
LOWERCARBON CAPITAL
foundation capital
SUMITOMO ELECTRIC
LEFT LANE
BILL & MELINDA GATES foundation
ALBUM VC
GROKVENTURES
Adapt Ventures
British International Investment
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Source: Crunchbase, Pitchbook, Tracxn

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# 4 key tailwinds have emerged in the last few years which have buoyed the Indian residential solar growth

## A Customer Economics

### Payback period under 5 years

With declining cost of solar modules and increasing cost of grid electricity, customer economics have become extremely attractive with IRRs >20% and payback period under 5 years in most states of India.

Early 2020s

## B Favorable Policy & Easy Permits

### One nation, one policy, one portal

- Centralized and seamless Permit process via a National Solar Portal
- First country in the world to make net-metering for homes a **legal right**
- Subsidies to consumers via Direct Bank Transfer
- Standardized policy and online process for permits across India

National Portal for Rooftop Solar

2022

## C Financing

### Availability of Solar Financing

Low interest rates (<11%) and long tenure (10 years) loans for residential solar from PSU banks.

SBI

pnb

2023

## D Consumer Awareness

### Consumer Awareness

Prime Minister advocating solar on homes as means to get free electricity for Indians. $9Bn approved in Union budget for subsidies and consumer awareness of residential solar in India.

PM SURYA GHAR MUFT BIJLI YOJANA

2024

Source: PIB; PV magazine India; Expert interviews

# …and customer experience of purchasing a Solar PV system has also changed significantly

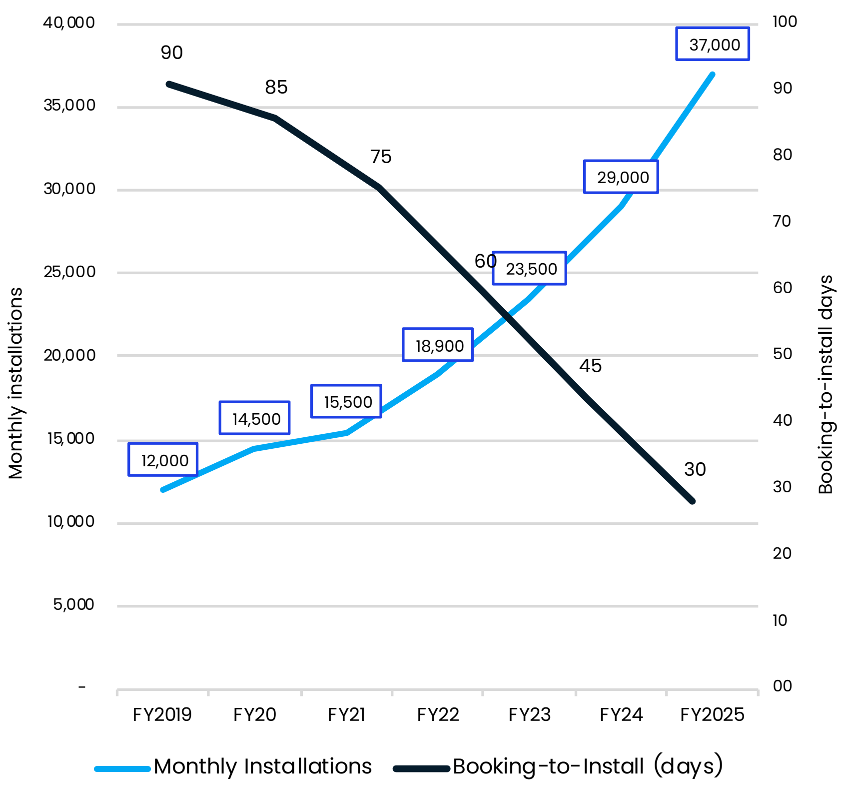
| | Degree of change |  Mr. Sharma (in 2020) |  Mrs. Mehta (in 2024) |
|---|---|---|---|
| **Subsidy disbursal** | ● | Claimed subsidy amount from State post installation; took >3 months | Direct benefit transfer within 30 days of installation; national policy & a unified platform |
| **DISCOM permit** | ◕ | Took more than 30 days and installer had to make 2-3 DISCOM visits for approval | Online, fast-tracked - Within 7 days |
| **Financing options** | ◕ | 5-yr. NBFC loan at 15 – 17%, high EMI | Loan from PSU bank at 11%; 10-year tenor |
| **Booking-to-installation time** | ◕ | 75 – 90 days | <45 days |
| **Breakeven/ Payback period** | ◑ | ~5 years | Less than 5 years |
| **Vendor options** | ◑ | Only Local independent mom ‘n’ pop shop | Option of a full-stack brand, OEM with channel dealer, etc. |

Source: Google reviews, Customer conversations

# Permit timeline and ease of application process has been a crucial factor in the rise of monthly installations

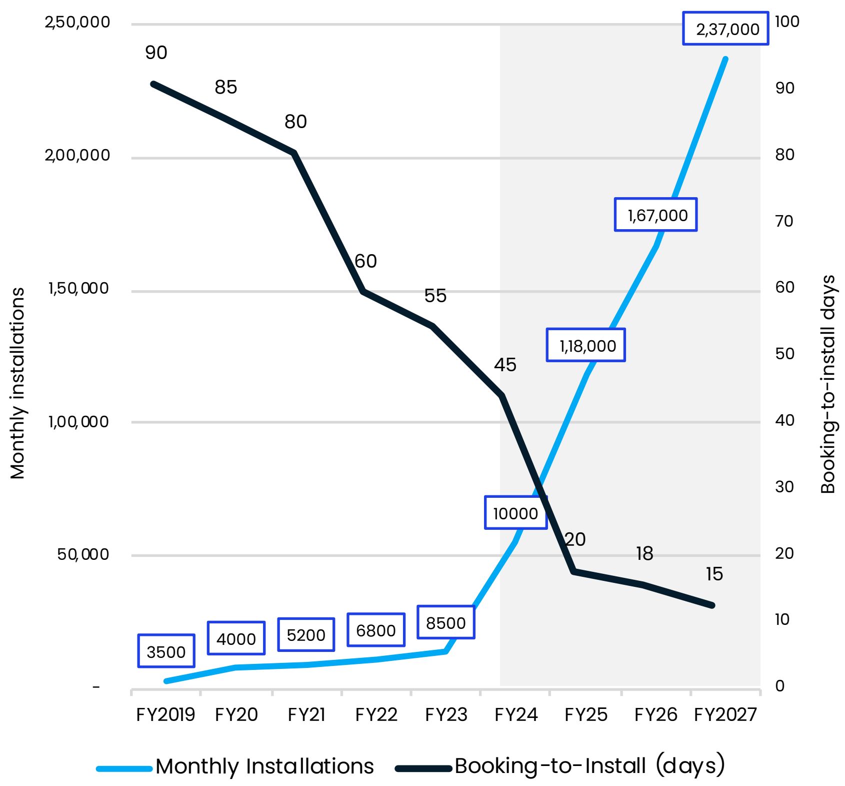
## By reducing approval days to 1/3rd, Germany has seen a 3x increase

Average # of days from booking-to-install




## Recent policy reforms are expected to drive a similar outcome in India

Average # of days from booking-to-install




## Driven by anchor solar policies in both countries

**Germany (EEG 2023 Reforms)**

- Introduction of a web portal for managing grid connections.
- Faster grid connection proceedings and tax simplifications.
- Streamlined application processes leading to a reduction in average permit timelines from 90 days in 2019 to 30 days in 2024.

**India (National RTS Portal 2024)**

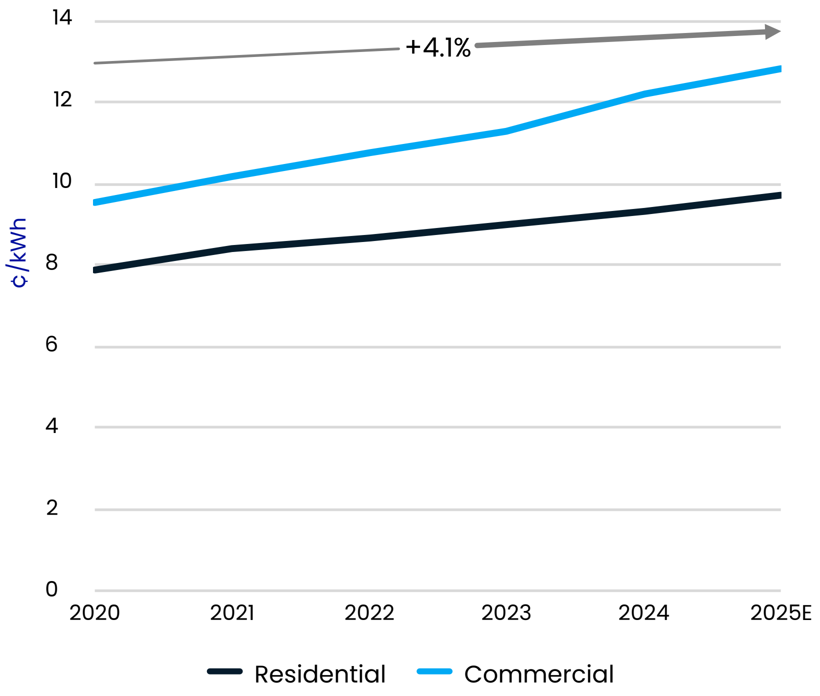
- The portal has integrated services from various stakeholders, including DISCOMs and state nodal agencies, ensuring a seamless and faster approval process.
- Direct Benefit Transfer (DBT) of the subsidy amount to customer's account within 30 days

Source: IEA, Ministry of Power, MNRE, Industry reports

# Residential solar economics become even more attractive as system cost declines and utility rates increase

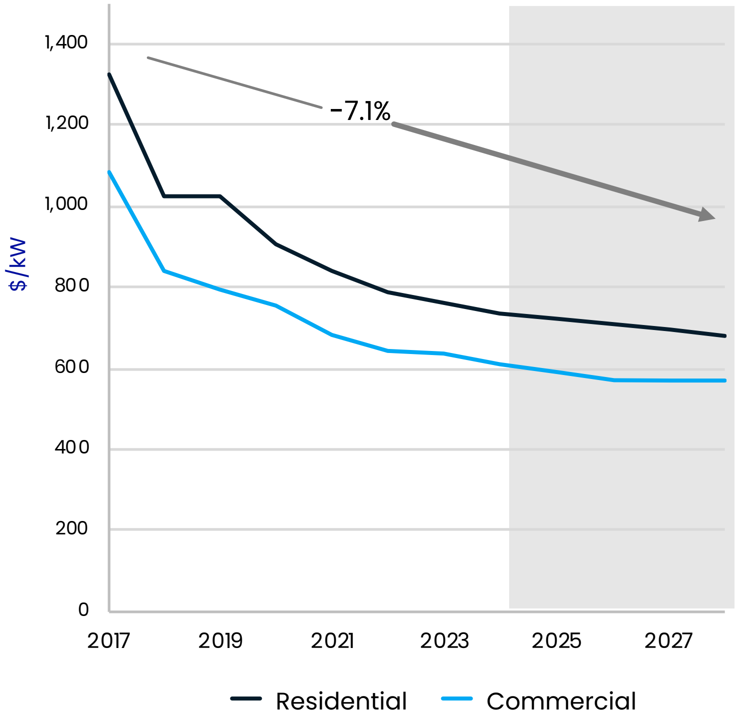
## Utility rates are going up as grid upgrades

India electricity prices (¢/kWh)[1]




## DER costs are declining1

Solar System Costs ($/kW)[2]




## Driven by many sustainable factors

**Customer demand for more control and lower cost options**

- Awareness of value proposition
- Levelized cost of PV reached grid parity for many segments, enabling attractive business cases
- Desire for reduced grid dependence/backup power

**Favorable regulations**

- Higher demand for DER products leads to scale effects in production

**Increasing benefits for customer from new technologies** to control loads and understand energy use

- Home energy monitoring
- Improved EMS systems
- Deeper smart meter penetration

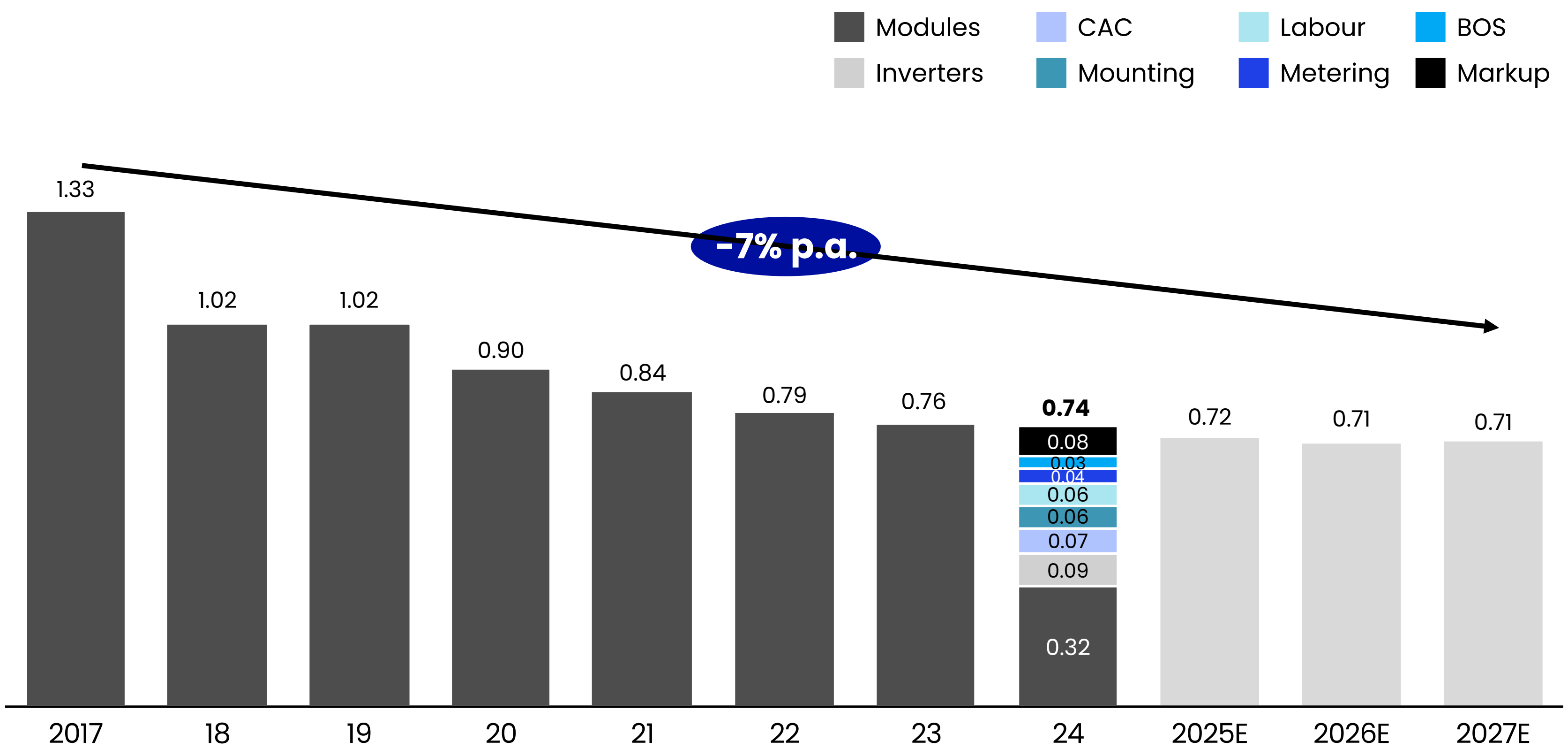
1. Wholesale costs
2. Cost of equipment and installation, including PV module, inverter and BOS, excludes installer margin

Source: IEA, Ministry of Power, MNRE, Industry reports

# Customer costs have decreased by ~7% p.a. from $1.33/W in 2017 to ~ $0.74/W in 2024

## Total customer cost for residential solar

$/W




1. 2017-2024 actual values are triangulated based on industry reports and government data to account for variation in samples. YoY decrease in $/W is similar across sources.
2. Estimated based on primary research of residential installers and weighted to include high cost (e.g. Solar Square) and low cost (e.g. local dealer) installers based on current market share. Total value and breakdown triangulated with industry reports

Sources: MNRE, DownToEarth, and public company reports

### Drivers for decreases in residential PV price include:

- Improved module efficiency and reduced hardware costs for structural and electrical parts
- Lower permitting costs
- CAC disruption (e.g., drive towards more digital sales)

# Central Government Schemes have made solar energy more accessible and affordable for residential consumers

| | 2017 | 2019 | 2020 |
|---|---|---|---|
| **Scheme** | **Sustainable Rooftop Implementation for Solar Transfiguration of India (SRISTI)** | **PM-KUSUM Scheme (Component C)** | **Grid-Connected Rooftop Solar Programme (Phase II)** |
| **Objective** | • To facilitate easy adoption of RTS by simplifying the application, approval, and installation processes | • The scheme aims to solarize pumps with a capacity of up to 7.5 HP and provide energy security to farmers | • To achieve a cumulative capacity of 40 GW from rooftop solar projects by 2022 |
| **Key Features** | • Streamlined procedures for subsidy disbursement.<br>• Online portals for application submission and monitoring.<br>• Standardized guidelines for installation and maintenance | • Solarize existing grid-connected agriculture pumps to ensure reliable and sustainable energy for irrigation purposes | • The scheme promotes residential solar adoption by reducing the upfront cost burden on consumers<br>• Provides performance-based incentives to DISCOMs to enhance rooftop solar installations |
| **Incentives** | • CFA covering a significant portion of the installation cost for residential, institutional, and social sector buildings.<br>• Additional incentives for DISCOMs for achieving solar rooftop installation targets. | • 30% of the cost being provided by the Central Government as a subsidy.<br>• 30% of the cost being provided by the State Government as a subsidy.<br>• The remaining 40% of the cost to be borne by the farmer/bank financing | • Central financial assistance (CFA) of up to 40% for rooftop solar systems up to 3 kW capacity.<br>• 20% CFA for systems between 3 kW and 10 kW.<br>• For group housing societies/residential welfare associations, 20% CFA is provided for systems up to 500 kW capacity. |

Sources: MNRE, PIB

# PM-Surya Ghar: Muft Bijli Yojana

## PM's residential solar plan is a game-changer

**A** The PM-Surya Ghar: Muft Bijli Yojana is a scheme launched by the Indian government to **promote the adoption of rooftop solar installations** for households.

**B** Announced by Prime Minister on Feb 13, 2024, the scheme aims to **enhance energy security** and sustainability through systemic reforms

### Key Features

**Systemic Reforms in RTS installation:** Shift to a unified permit processes via a centralized National portal, along with availability of financing options

**Total Outlay:** The scheme has a total financial outlay of ₹75,201 crore

**Streamlined Regulatory Approvals:** Deemed net-metering approvals across India for easy integration

### Implementation

**Target Beneficiaries:** The scheme targets 1 Cr households, particularly focusing on those in rural and semi-urban areas to improve energy access and affordability

**National Portal for Rooftop Solar:** A streamlined and user-friendly application process through the official portal, making it easy for households to apply and benefit from the scheme

### Highlights

**1,00,00,000+**
Target RTS installations in 3 years

**10.5-11% Interest Loans**
Making solar accessible through PSU banks.

**$9 Bn**
Approved Financial Outlay of the scheme

**$80 Mn**
Budgeted for awareness campaigns

Sources: MNRE, PIB, PM Surya Ghar Program Office

# Consumer interest in resi-solar has exploded since the announcement of the program

**10L+**

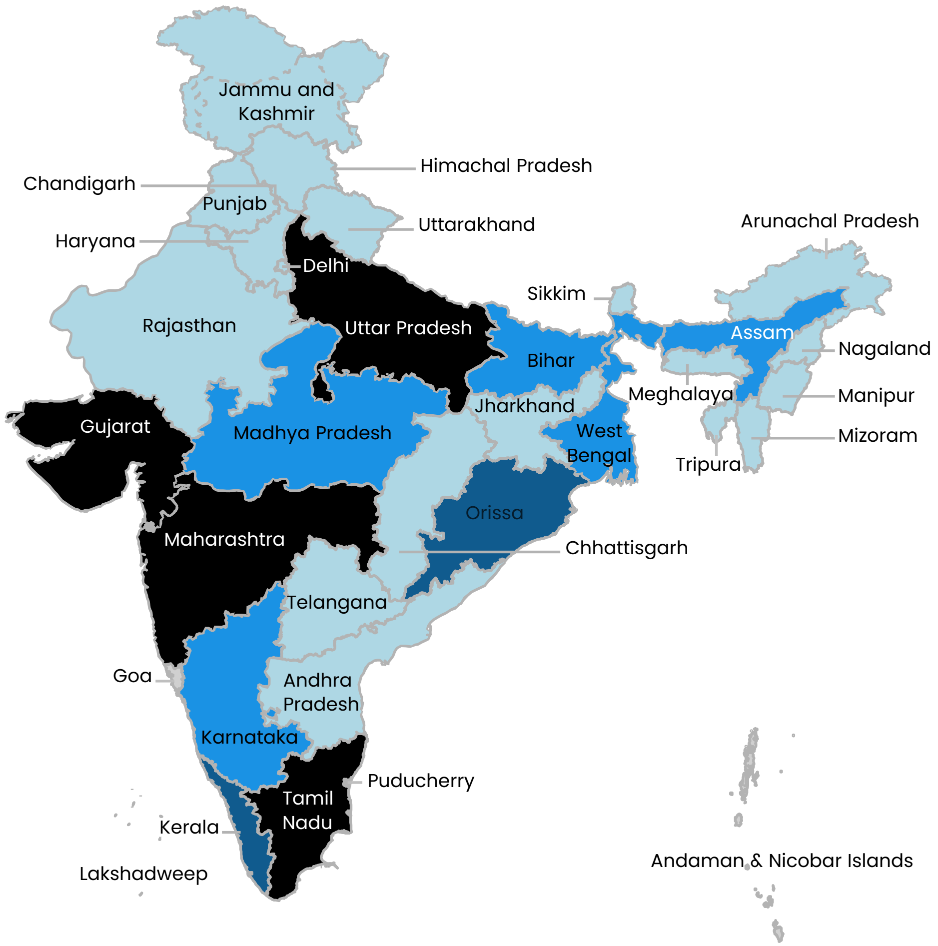
applications received under the plan

**2.3L+**

installed in <5 months

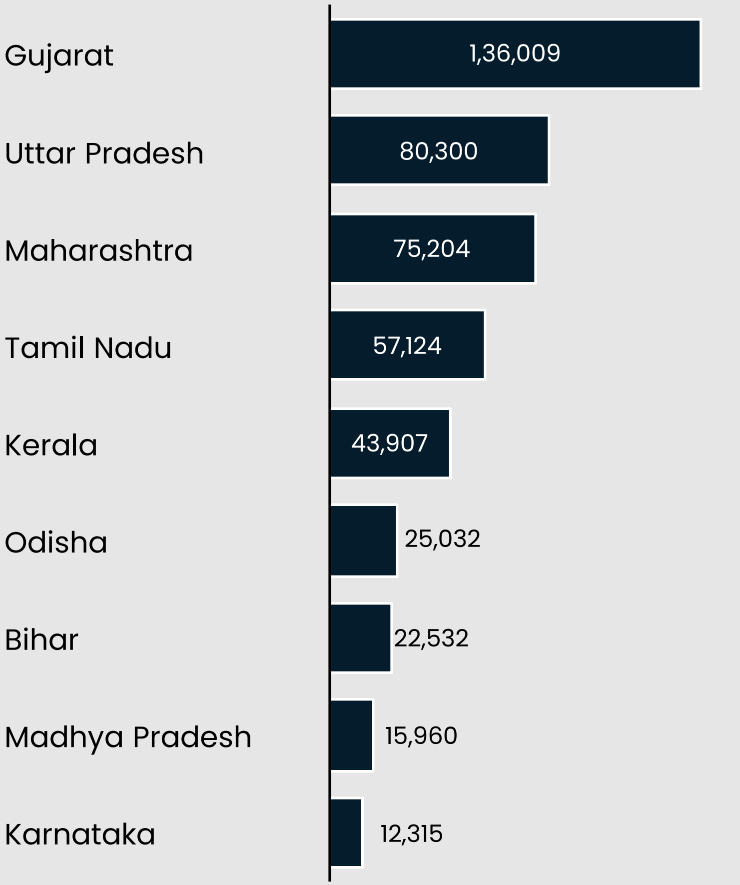
**State wise resi-solar applications in first 4 months of launch**

<5k | 5k-10k | 10k-25k | 25k-50k | >50k




Total Application

**Total applications since program launch[1]**




1. Data as on July 17, 2024

Source: PM Surya Ghar Program Office

# The program represents a transition from a state-subsidized initiative to a centralized program with a unified platform

**Unified Platform**

- The portal serves as a centralized platform for all stakeholders involved in the rooftop solar process, including homeowners, DISCOMs, state nodal agencies, and installers.
- It integrates various state and central schemes to provide uniform information and application processes

**Online Application Submission**

- Homeowners can submit their applications for rooftop solar installations through the portal.
- The portal provides a step-by-step guide on how to fill out the application, what documents are needed, and how to upload them.

**Document Verification**

- The portal facilitates the online verification of submitted documents.
- Automated systems check the completeness and accuracy of the application, reducing the time needed for initial approval

**Real-Time Tracking and Updates**

- Applicants can track the status of their application in real-time.
- The portal provides updates at each stage of the process via email and SMS notifications, ensuring transparency and keeping homeowners informed

**Integration with DISCOMs and State Agencies**

- The portal is integrated with various DISCOMs and state nodal agencies, ensuring that applications are forwarded to the appropriate authorities seamlessly.
- This integration helps in coordinating inspections, approvals, and other necessary steps efficiently

**Subsidy Calculation and Disbursement**

- The portal calculates the eligible subsidy for each application based on the guidelines set by MNRE.
- Once approved, the subsidy is directly transferred to the homeowner’s bank account, making the financial aspect straightforward and transparent

**Vendor Management**

- The portal lists registered vendors who are authorized to install rooftop solar systems. Homeowners can select vendors, view their offerings, and compare prices

**Virtual Inspections and Approvals**

- The portal supports virtual inspections, where necessary, to expedite the approval process.
- Digital agreements can be signed online, further reducing the need for physical visits

Source: PIB, PM Surya Ghar website, National Portal for Rooftop Solar

# Projected growth is concentrated in mature markets; states with strong economics (e.g. MH, UP, GJ) will grow quickly

Increase in the installed resi solar capacity p.a. 2020 to 2030

**Cumulative forecasted 2020-2030 capacity,** *GW*

<1GW | 1-2GW | 2-3GW | 3-5GW | 5GW+


$

**2024E payback period (yrs[1])**

| | |
|---|---|
| Maharashtra | 3-4 yrs |
| Kerala | 4-6 yrs |
| Gujarat | 4-5 yrs |
| Rajasthan | 4-6 yrs |
| Uttar Pradesh | 4-6 yrs |
| Madhya Pradesh | 4-6 yrs |
| Haryana | 5-6 yrs |
| Telangana | 5-7 yrs |
| Karnataka | 5-7 yrs |
| Tamil Nadu | 5-7 yrs |

1. Estimated for an average monthly household consumption of 350 kWh

Source: MNRE, Company estimates

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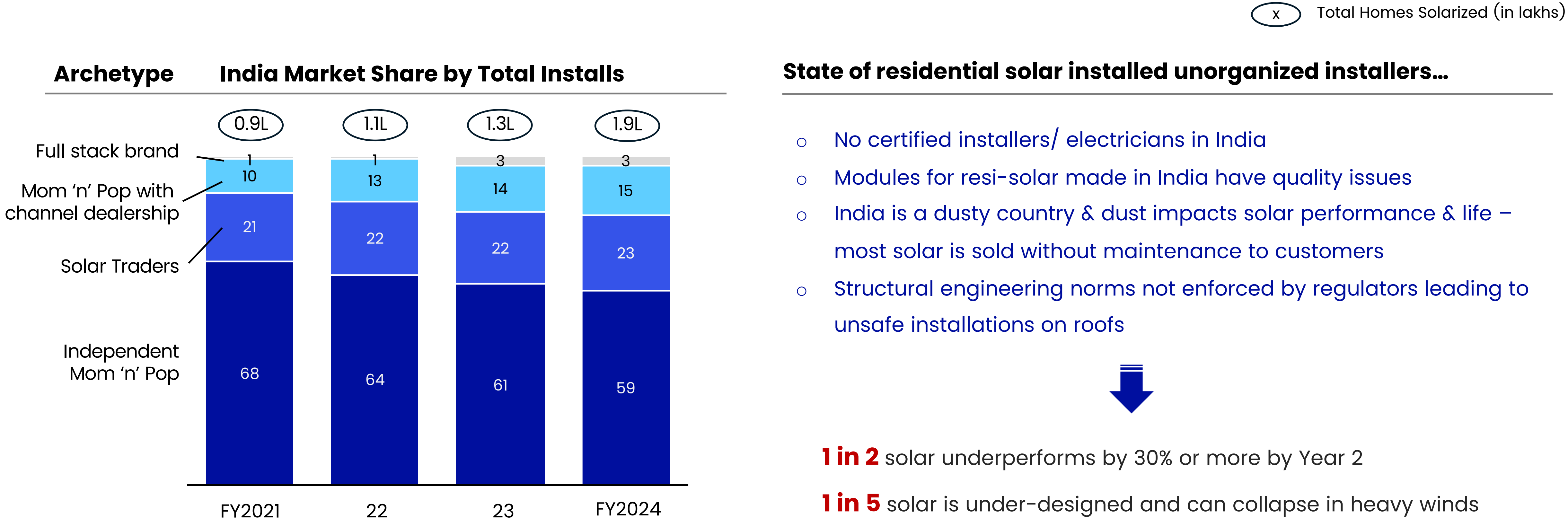
# Nationally, the residential solar market is highly fragmented, dominated by 3 archetypes of local players…




**OEMs with a dealer network**

## State of residential solar installed unorganized installers…

- No certified installers/ electricians in India
- Modules for resi-solar made in India have quality issues
- India is a dusty country & dust impacts solar performance & life – most solar is sold without maintenance to customers
- Structural engineering norms not enforced by regulators leading to unsafe installations on roofs

**1 in 2** solar underperforms by 30% or more by Year 2

**1 in 5** solar is under-designed and can collapse in heavy winds

**1 in 5** solar is sold via middle-men with incentive misalignment

**1 in 20** panels on field have manufacturing/ handling damages

**1 in 2** panels warranty claims are rejected leading to customer's loss

Source: MNRE, MERCOM, CEEW research of residential solar performance, primary research with module OEMs, Company estimates

# …who operate at different scales with different margin profiles

| | Full stack brand | Independent Mom 'n' Pop | Mom 'n' Pop with dealership | Solar Traders |
|---|---|---|---|---|
| Key features | Fully vertically integrated across the value chain.<br><br>Direct to consumer with no middle-men.<br><br>Operating nationally, with 50km radius in each city.<br><br>Dense customer network.<br>Offer after-sales.<br>Offer point of sale financing.<br>Brand promises.<br><br>Value led sales, not price led. | Local operators working in single city within a 30-50 km radius.<br><br>Most try to play in >5 kW customer segment, to maximize profit per order.<br><br>5-10 member teams.<br>Most struggle to offer after-sales.<br>Struggle to service 3-4 kW customers (85% of India)<br>Low safety and quality standards are.<br><br>Pricing and relationship led sales. | Local operators with working capital to stock inventory which allows them to take an OEM dealership. Single city businesses, operate in 30-50 km radius<br><br>Many are primarily B2B distributors of solar products, but also do solar installation for end consumers. Non standardized last mile installation, pricing, service and quality.<br><br>5-10 member teams.<br>Most struggle to offer after-sales.<br><br>OEM brand halo led sales | Mostly part-time solar installers. Locally operate in a 5-10 km radius of shop.<br><br>Hire local daily wagers to install. Mostly amateur entrepreneurs, selling within close network/ locality. Procure materials on order-to-order basis from OEM dealers.<br><br>2-5 member teams.<br>Don't offer after-sales maintenance.<br>Don't have tie-ups to give point of sale financing.<br><br>Cheapest price led sales. |
| Market trend | ↑ Counter positioning with brand promises like savings guarantee. Driving market consolidation.<br><br>Benefit from #1 rank on India's national solar portal. | → Play on lower price and low quality. 20-30% of these operators churn out of industry every year. Top 50 regional incumbents in India have local city reputation and reach scale of 50+ homes per month. | ↘ Benefit from brand halo. But higher price then other mom & pop operators, nonstandard offering and less value prop to customers. | → Fly by night operators. 30-40% of these operators churn out of industry every year. Play on cheapest price by selling low quality and operating on very low overheads. |
| # of dealers/shops | 1 | ~2000 | ~2000 | ~3000 |
| ~Monthly installs per dealer | ~800 | 90% install <10 homes a month | 90% install <10 homes a month | 90% install <5 homes a month |
| Net Margin per home | 37k per home | 25-40k per home | 25-40k per home | 5-10k per home |

Source: Expert Interviews, Press search

# 50%+

solar systems installed by local mom ‘n’ pop underperform by 30% or more by Year 2

## …and a majority of them have flooded the market with sub-standard quality solutions

**80%**

suffer from soiling loss and lack of solar maintenance

**50%**

Suffer from poorly engineered solar often installed in shadows

**30%**

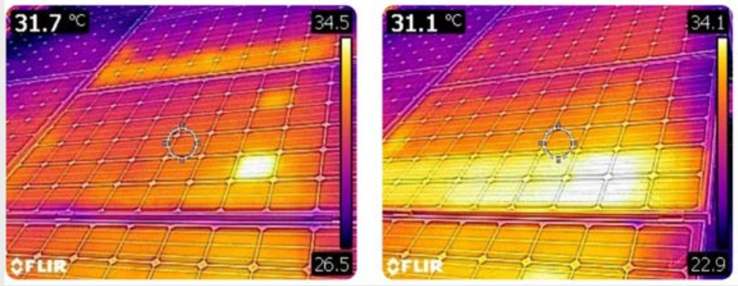
Suffer from module defects

**12%**

Face other issues such as scaling, snail trials, hotspots etc.

*Irradiation loss due to dust deposition*

*Low energy production due to shading*




*Hotspots due to defective/cracked cell*

# Every customer will be a first-time solar buyer, and solar will be the 3rd biggest purchase of their life (~3000$+)…

## Persona of a typical Solar customer..

- <$15,000 p.a. Family Income
- $400-800 p.a. electricity bill
- 70% are 2W families, don’t own a car
- <15% own a air-conditioner
- First time solar buyers
- Value conscious

## ..and their roadblocks in going Solar

**A** Lack of awareness & misconceptions

*“How much solar do I need? Does it work in monsoons? Does it need maintenance?”*

**B** Uncertainty of ROI

*“How much exactly will I save? Is there any risk of power production?”*

**C** Complexity of the installation process

*“Is it too much hassle?  How long will the govt. permits take? Will I struggle to get my subsidy?”*

**D** Affordability

*”Can I afford the upfront investment?”*

# Solar customers in India can broadly be categorized using this 2x2 framework

**Not exhaustive**

| Attitude towards Rooftop Solar | Solar Plant capacity: High (>3 kW) | Solar Plant capacity: Low (<=3 kW) |
|---|---|---|
| **Reactive** (Solar system is a reaction to rising power bills) | **Unstinted Indulgers** | **Striving Sustainers** |
| **Proactive** (Manageable power bill but Solar is an investment for future) | **Savvy Planners** | **Cautious Conservatives** |

# Resi-solar customer archetypes (1/2)

**Not exhaustive**

| |  **Striving Sustainer (Classic Middle Class)** Reactive Purchase, Low Capacity Installation |  **Savvy Planner (Well educated professionals)** Proactive Purchase, High Capacity Installation |
|---|---|---|
| **Key characteristics** | • Aged 30 to 50, found in both metros and non-metros<br>• Have a basic level of education and are employed either in an organization at a middle level or have their own business | • Aged from mid 30s to late 40s, married with young kids, mostly in Metro and Tier-1 towns<br>• Well-settled professionals or well-to-do entrepreneurs, and aware of Solar as a sustainable solution. |
| **Energy consumption/ Need for Solar** | Limited power guzzling appliances, increasing electricity bill or a step change in consumption derails their monthly budget. Peak summer/winter bills puts a dent on finances | Have a general interest in and awareness of new trends, already know the basics of home solar. Those constructing a new home, see it as a good opportunity to install a solar plant as installation won't cause major disruption |
| **Purchase Journey** | Approach solar from a 'Savings' mindset and keenly follow subsidy schemes. Decision is made mostly by highest paying family member. **Loan, EMIs and 'limited period discounts' are a definite attraction** | **Willing to place a premium on professionalism** and convenience. They gauge entities by its credentials, reviews and mannerisms of sales professionals. In case of those constructing a new home, architect or contractor could become an important gatekeeper. |
| **Watchouts** | They take time to make their decision and at times lack required roof size & space. **Easily swayed by the pitch of a local player who offers low cost** | They would also **focus on aspects like installation design and predictable service experience**. They also have a bias towards known brands like TATA etc. that they know from research |

# Resi-solar customer archetypes (2/2)

**Not exhaustive**

| |  **Cautious Conservatives (Retirees/Semi-retirees)**<br>Proactive Purchase, Low Capacity Installation |  **Unstinted Indulgers (India's 1%)**<br>Reactive Purchase, High Capacity Installation |
|---|---|---|
| **Key characteristics** | ▪ Retirees or semi-retirees living in an empty nester home<br>▪ Middle to upper middle class lifestyle; often have a son or a daughter who is well-educated and working in another city<br>▪ Value conscious | • High-income senior professionals and businessmen, in metro and tier-1 towns, in their late 40s and beyond<br>• Includes doctors, lawyers, govt. officials, PSU managers, etc, |
| **Energy consumption/ Need for Solar** | Emotional about their home, they've realized some gaps when kids recently came to stay over during pandemic and look forward to bridge them. Motivated to explore Solar driven by advice of their kids or those who've already installed it | Multiple heavy-duty power guzzling appliances. Unstinted usage of these appliances with multiple people using them in parallel. They seek a solution that would help them continue with their desired lifestyle with lower electricity bills. |
| **Purchase Journey** | Their **primary source of awareness is WOM** + vendor spiel. Their decision making is a joint one - involving husband, wife, son/daughter as well as a local perceived expert (who can be someone who installed solar plants in other homes or some local electrician/person with technical expertise). | Have discovered Solar as a solution through close-circle WOM. They want an assurance of getting the best ROI and do their groundwork before approaching the players. They would **expect their installation partner to handle things predictably** with minimal hiccups and a robust post-purchase assistance. |
| **Watchouts** | Rely on local WOM, experts and relationships. Given their long-standing relationships in their locality, they **expect on-call service that can resolve their queries** and handhold them | They **could become a strong evangelist of the entity** and would set up a virtuous cycle. Building relationship with them requires convincing them that they are talking to someone 'senior' who treats them as a high value priority client |

## Contents



# Disruptor Case Examples

- **Deep-Dive: Enpal**
- Deep-Dive: 1KOMMA5°

# ENPAL is Germany's #1 resi-solar player, capturing ~9% market share with a leasing business model

Enpal.

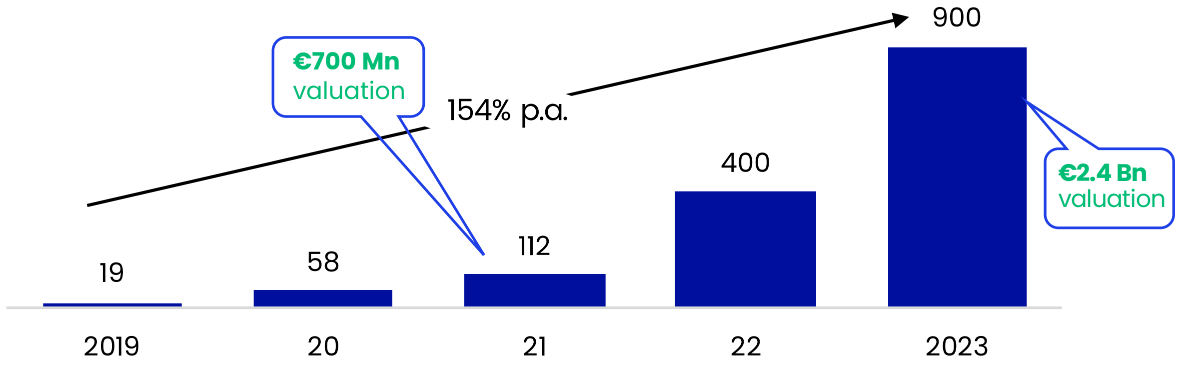
## Business description

ENPAL offers **leasing (>90% of sales) of solar panels** and additional equipment (e.g. batteries) to private homeowners.
**Less than 10% of sales through direct sales**

**Investors:**

SoftBank Vision Fund
HV CAPITAL
PRINCEVILLE CAPITAL
TPG RISE CLIMATE
THE WESTLY GROUP

## Financial highlights, revenues in € Mn




## Geographic footprint

ENPAL´s core market
ENPAL´s new markets

ENPAL generates >85% of its revenues in Germany – current market leader

ENPAL is expanding its presence to Italy

## Highlights

**€645 Mn**
Total equity funding raised till date

**€2.4 Bn**
Last round valuation (revenue of ~€900 Mn)

**3,400+**
Employees including ~1.5k installers and craftsmen

**3,000+**
Monthly PV system installations

Source: Crunchbase, Tracxn, Founder interviews, Press search

# Founded in 2017, ENPAL has achieve a strong growth trajectory, raising a €215mn Series D in 2023

## ENPAL history (2017-23)

Funding events | Operational events

### 2017

**2017:** ENPAL is founded by Mario Kohle and Viktor Wingert, with a mission to provide affordable and accessible renewable energy to homeowners in Germany.

### 2019

**2019:** The company raised an undisclosed amount of seed funding from Picus Capital, Lukasz Gadowski and Alexander Samwer

**2019:** ENPAL launches its leasing model, which allows homeowners to install solar panels on their homes without any upfront costs.

### 2020

**2020:** ENPAL raised €21mn in Series A venture funding from Picus Capital and Lukasz Gadowski

**2020:** The company received €120mn of debt financing in the form of a loan from undisclosed lenders

### 2021

**2021:** ENPAL finalized a debt refinancing round for €345 mn. BlackRock, Pricoa Private Capital, and UniCredit contributed €275 mn senior secured debt, while undisclosed lenders provided a subordinated mezzanine loan worth €70mn

**2021:** The company raised €250mn of Series C funding led by HV Capital and SoftBank Vision Fund 2

### 2022

**2022:** ENPAL has raised €855mn of committed debt from funds and accounts managed by Blackrock Alternatives, ING, Pricoa Private Capital, Unicredit

**2022:** ENPAL opens a training center for solar installers and electricians in Blankenfelde-Mahlow in the state of Brandenburg

**2022:** ENPAL received a Junior Debt financing commitment of up to €70mn from Prime Capital AG

### 2023

**2023:** ENPAL has secured €215mn in Series D funding led by TPG Rise Climate, with participation from Westly Group, Activate Capital, HV Capital, SoftBank Vision Fund 2, and Princeville Climate Technology

**2023:** ENPAL secures €356mn debt commitments from Citi and M&G investments for first German residential solar securitization

Source: Company website, Press research, Pitchbook

# Disruptor Case Examples

- Deep-Dive: Enpal
- **Deep-Dive: 1KOMMA5°**

# Founded in 2021, 1KOMMA5° generates €460 Mn+ in sales – it pursues an aggressive consolidation strategy to grow

## Business description

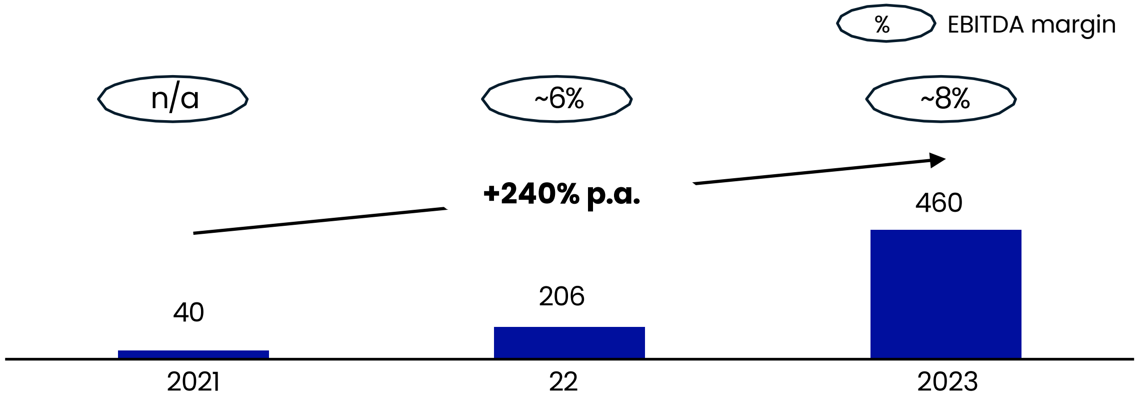
1KOMMA5° **offers all-in-one solutions** around resi solar PV systems combining it with heat pumps, battery storages, wall boxes, energy mgmt. solutions

Core strengths: **Complementary and innovative hardware systems** together with having **own qualified installers** ensuring fast installation

**Investors:**

G2 VENTURE PARTNERS, <norrsken>, Hamilton Lane, eCAPITAL ENTREPRENEURIAL PARTNERS

## Financial highlights, revenues in € Mn




## Geographic footprint

| | |
|---|---|
| **Nordics** | 9 |
| **Germany** | 42 |
| **Italy** | 9 |
| **Australia** | 6 |

Aggressive consolidation strategy has helped the company expand rapidly

Currently 1KOMMA5° has almost 60 locations across 5 countries

## Highlights

**€502 Mn**
Total equity funding raised till date

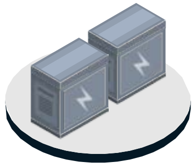
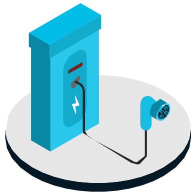
**€1.6 Bn+**
Estimated current valuation

**1,300+**
Employees including installers and craftsmen

**100,000+**
PV system installations so far

Source: Crunchbase, Tracxn, Founder interviews, Press search

# 1KOMMA5° offers a range of products to create a complete energy system

| Products | | Description |
|---|---|---|
|  | **Solar panels** | 1KOMMA5° **mainly focuses on the sale of soar systems**<br>Their **offering includes personal consulting, installation and maintenance**<br>The company gives its customers **25 years guarantee** on the products |
|  | **Energy storage** | Adjacent to its solar panels, the company also **sells and installs energy storages**<br>Combined with the Heartbeat app this allows for an **optimized energy consumption** |
|  | **Wallbox** | Next to solar systems 1KOMMA5° offers **consultation and installation for wallboxes** which can be directly connected to all other products |
|  | **Heat pumps** | Next to solar systems 1KOMMA5° offers **consultation and installation for heat pumps** which can be directly connected to all other products |

## Key insights

1KOMMA5° offer covers the **full range of hardware energy products to build a complete energy system** which can be connected to their **proprietary "Heartbeat" energy mgmt. application**

The company markets its offer-ings as a **one-stop-solution with a high-quality standard and broad range of services**

In contrast to other providers the **company only offers cash purchases and no leasing**

## OEM partnerships

sonnen AEROCOMPACT® ENPHASE. MEYER BURGER

Source: Company website; Press

# Resi-solar has multiple success stories across geographies, backed by a global pool of investors

| Company | HQ | Company Overview | Funds raised | Equity Investors | Revenues (2023) |
|---|---|---|---|---|---|
| Enpal | Germany | Provides solar leasing options for residential customers. | €645 million | SoftBank Vision Fund, HV Capital, TPG Rise Climate | €900 million |
| 1KOMMA5° | Germany | Focuses on building a one-stop-shop for solar installations and energy management systems. | €502 million | G2 Venture Partners, Porsche Ventures, <norrsken> | €460 million |
| Brighte | Australia | Provides financing solutions for residential solar installations. | $207 million | Grok Ventures, Qualgro, AirTree | $25 million |
| solfácil | Brazil | Offers financing and installation of solar panels for residential and commercial properties. | $165 million | QED Investors, SoftBank Group, Valor Capital Group, Vinci Partners | $20 million |
| goodleap | US | Provides financing solutions for residential solar and other sustainable home improvements. | $800 million | Goldman Sachs, TPG | $200 million |
| gosolr | South Africa | Offers a subscription-based model for solar installations, including maintenance and monitoring. | $50 million | ARC Investments | $10 million |
| Project Solar | US | Specializes in residential solar installations, focusing on affordability and efficiency. | $23 million | Left Lane, philo, Album VC | - |
| octopus energy | UK | Renewable energy provider known for competitive tariffs and innovative customer service. | $2.84 billion | generation, CPP Investments, Galvanize Climate Solutions, lightrock, Macquarie | $16 billion (FY 2023) |
| M-KOPA | Kenya | Provides solar home systems to off-grid customers in Africa through a pay-as-you-go (PAYG) financing model. | $245 million | generation, Shell Foundation, Lightstone, IFC, BII, lightrock, Sumitomo Electric, CDC, Bill & Melinda Gates foundation | $472 million (estimate) |
| niko | Mexico | Niko is a Mexico-based provider of residential solar electricity and solar power services for homeowners. | $3.3 million | 468 Capital, Adapt Ventures, Picus Capital | - |
| sunrun | US | Leading provider of residential solar, battery storage, and energy services in the US. | $1.8 billion | Accel Partners, foundation capital, idinvest Partners, de Anda Capital, Sequoia, Investec | $2.2 billion (CY 2023) |
| vivint.Solar | US | Provides residential solar installations and services in the US. | Acquired | Summit Partners, Blackstone, Valor, Brookfield | $450 million (TTM as on day of delisting) |

Source: Crunchbase, Pitchbook, Tracxn

# Thank You